

COURSE TITLE : **DATA COMMUNICATION**
COURSE CODE : **3151**
COURSE CATEGORY : **B**
PERIODS/WEEK : **4**
PERIODS/SEMESTER : **60**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Concepts of Data Communication	15
2	Physical Layer	15
3	Transmission Media	15
4	Data Link Layer	15

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	Understand the concept of Data Communication
	2	Understand the concept of networks
	3	Understand the concept of network model
2	1	Understand Data and Signals
	2	Understand Digital and Analog Transmission
	3	Understand Bandwidth Utilization -Multiplexing
3	1	Understand Guided and Unguided transmission media
	2	Understand Switched Networks.
4	1	Understand Error Detection and Correction
	2	Understand Data Link Controls

Specific Outcomes:

Module I. Overview of Data Communication and Networking.

1.1 Understand the concept of Data Communication.

- 1.1.1 Explain data communication concepts.
- 1.1.2 Explain briefly components of data communication.
- 1.1.3 Discuss different data representation forms.
- 1.1.4 Discuss different data flow methods.

1.2 Understand the concept of networks.

- 1.2.1 Define different network attributes.
- 1.2.2 Discuss different physical structures of network.
- 1.2.3 Discuss categories of networks.
- 1.2.4 Explain interconnection of Networks.
- 1.2.5 Define protocol.
- 1.2.6 Discuss various standards.

1.3 Understand the concept of network model

- 1.3.1 Discuss the layered approach.
- 1.3.2 Explain ISO OSI layered architecture.
- 1.3.3 Describe the functions of layers in OSI model.

Module II. The Physical Layer.

2.1 Understand Data and Signals.

2.1.1 Differentiate Analog and Digital Data and signals.

2.1.2 Explain Periodic analog signals.

2.1.3 Define the terms sine wave, Phase, Wave length, time and frequency domains, and bandwidth.

2.1.4 Explain Digital signals.

2.1.5 Define the terms bit rate and bit length.

2.1.6 Describe digital signals transmission.

2.1.7 Describe Transmission impairments and various types.

2.2 Understand Digital and Analog Transmission

2.2.1 Explain Analog to Digital Conversion

2.2.2 Explain Transmission modes

2.2.3 Explain Digital to Analog Conversion

2.2.4 Explain Analog to Analog Conversion.

2.3 Understand Bandwidth Utilization -Multiplexing

2.3.1 Explain Frequency Division Multiplexing.

2.3.2 Explain Wavelength Division Multiplexing.

2.3.3 Explain Time Division Multiplexing.

Module III. The Transmission Media.

3.1 Understand Guided transmission media.

3.1.1 Explain about Twisted pair cable.

3.1.2 Explain about Coaxial Cable.

3.1.3 Explain about Optical fiber cables.

3.2 Understand Unguided media.

3.2.1 Describe wireless propagation.

3.2.2 Describe Radio Waves.

3.2.3 Describe Micro Waves.

3.2.4 Describe Infrared Waves.

3.3 Understand Switched Networks.

3.3.1 Explain Circuit switched Networks.

3.3.2 Explain Packet switching – datagram, virtual circuit.

3.3.3 Explain the structure of a switch.

Module IV. Data Link Layer Services

4.1 Understand Error Detection and Correction

4.1.1. Define Key terms related to error detection and correction.

4.1.2. Explain different coding schemes

4.1.3. Discuss block codes

4.1.4. Discuss cyclic codes

4.1.5. Explain error detection and correction methods

4.1.6. Explain check sum error detection

4.1.7. Explain forward error correction methods

4.2 Understand Data Link Controls

4.2.1. Explain the framing concepts

4.2.2. Discuss flow and error control

4.2.3 Discuss Data link layer protocols

4.2.4 Explain HDLC

4.2.5 Explain point to point protocol

4.3 Understand random access protocols like ALOHA, CSMA, CSMA/CD

CONTENT DETAILS

MODULE I – CONCEPTS OF DATA COMMUNICATION

Concepts – components of communication - representation- text, numbers, images, audio, video – Data flow – simplex, half duplex, full duplex – Network attributes – performance, reliability, security – Physical structure – type of connections, topology – Categories – LAN, WAN, MAN – Inter connection - circuit, packet – protocols – standards – layered approach – ISO OSI model – functions of layers.

MODULE II – PHYSICAL LAYER

Analog and digital – data, signals – Periodic analog signals – sine wave, phase, wave length, time and frequency domains, bandwidth – digital signals – bit rate, bit length – digital transmission – base band, broad band – impairments – attenuation, distortion, noise – analog to digital – PCM– Transmission modes – parallel, serial , asynchronous, synchronous – digital to analog – ASK, FSK, PSK – analog to analog – AM, FM, PM – Multiplexing – FDM, WDM, TDM

MODULE III – TRANSMISSION MEDIA

Guided – twisted pair, co-axial, fiber optic – unguided – wireless – radio, micro wave, infrared – switched networks – circuit switched – packet switched – datagram, virtual circuit – switch – structure – space division – time division – structure of packet switch

MODULE IV – DATA LINK LAYER SERVICES

Error detection and correction – terms – types of errors, redundancy, detection versus correction, coding – coding schemes – block code , parity check – cyclic codes – CRC– check sum – forward error correction – framing – character oriented, bit oriented – flow and error control – DLL protocols – Simple, Stop and wait – HDLC – Point to point protocol– ALOHA, CSMA, CSMA/CD.

Text Book(s):

1 . Data Communications and Networking – Behrouz A. Forouzan – McGraw Hill Edn.-Fourth Edition/Fifth Edition

REFERENCES:

1. Computer Networks – Andrew S. Tanenbaum – Prentice Hall-Fifth Edition
2. Data Communication & Networks - William Stallings- Prentice Hall-Tenth Edition
3. Data Communications, Computer Networks and Open Systems –Fred Halsall , Addison-Wesley, 1996